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Vector scanning augment reality

Abstract

A method of displaying images by vector scanning in a head-mounted display includes determining, based on content of an image to be displayed by the head-mounted display, a vector scanning pattern for displaying the image, the vector scanning pattern including a plurality of scan lines and one or more transition lines connecting the plurality of scan lines, at least one transition line of the one or more transition lines including a curved section (or loop); generating control signals for controlling a scanner to steer, according to the vector scanning pattern, a light beam emitted by a light source towards a waveguide display of the head-mounted display; and generating modulation signals for modulating the light source such that the light beam is turned on when the scanner operates according to the plurality of scan lines and is turned off when the scanner operates according to the one or more transition lines.

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Background/Summary

BACKGROUND

(1) An artificial reality system, such as a head-mounted display (HMD) or heads-up display (HUD) system, generally includes a head-mounted display system in the form of a headset or a pair of glasses and configured to present content to a user via an electronic or optic display within, for example, about 10-20 mm in front of the user's eyes. The head-mounted display system may display virtual objects or combine images of real objects with virtual objects, as in virtual reality (VR), augmented reality (AR), or mixed reality (MR) applications. A near-eye display generally includes an optical system configured to form an image of a computer-generated image on an image plane. The optical system of the near-eye display may relay the image generated by an image source (e.g., a display panel) to create a virtual image that appears to be away from the image source and further than just a few centimeters away from the user's eyes.

SUMMARY

(2) This disclosure relates generally to near-eye display. More specifically, and without limitation, techniques disclosed herein relate to vector scanning-based near-eye display. Various inventive embodiments are described herein, including devices, systems, structures, methods, processes, program code, and the like.

(3) According to certain embodiments, a head-mounted display system may include a light source

configured to emit a light beam, a scanner configured to steer the light beam, a waveguide display configured to receive the light beam from the scanner and direct the light beam towards an eyebox of the head-mounted display system, one or more processors electrically coupled to the light source and the scanner, and one or more memory devices storing instructions. The instructions, when executed by the one or more processors, cause the one or more processors to perform operations including: determining, based on content of an image, a vector scanning pattern for displaying the image, where the vector scanning pattern includes a plurality of scan lines and one or more transition lines connecting the plurality of scan lines, and where at least one transition line of the one or more transition lines includes a curved section; generating control signals for controlling the scanner to steer the light beam according to the vector scanning pattern; and generating modulation signals for modulating the light source such that the light beam is turned on when the scanner operates according to the plurality of scan lines and is turned off when the scanner operates according to the one or more transition lines.

(4) According to certain embodiments, a processor-implemented method may include determining, based on content of an image to be displayed by a head-mounted display system, a vector scanning pattern for displaying the image, where the vector scanning pattern includes a plurality of scan lines and one or more transition lines connecting the plurality of scan lines, and where at least one transition line of the one or more transition lines includes a curved section; generating control signals for controlling a scanner to steer, according to the vector scanning pattern, a light beam emitted by a light source towards a waveguide display of the head-mounted display system; and generating modulation signals for modulating the light source such that the light beam is turned on when the scanner operates according to the plurality of scan lines and is turned off when the scanner operates according to the one or more transition lines.

(5) This summary is neither intended to identify key or essential features of the claimed subject matter, nor is it intended to be used in isolation to determine the scope of the claimed subject matter. The subject matter should be understood by reference to appropriate portions of the entire specification of this disclosure, any or all drawings, and each claim. The foregoing, together with other features and examples, will be described in more detail below in the following specification, claims, and accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Illustrative embodiments are described in detail below with reference to the following figures.
- (2) FIG. 1 is a simplified block diagram of an example of an artificial reality system environment including a near-eye display.
- (3) FIG. 2 is a perspective view of an example of a near-eye display in the form of a head-mounted display (HMD) device for implementing some of the examples disclosed herein.
- (4) FIG. 3 is a perspective view of an example of a near-eye display in the form of a pair of glasses for implementing some of the examples disclosed herein.
- (5) FIG. 4 illustrates an example of an optical see-through augmented reality system including a waveguide display.
- (6) FIG. 5A illustrates an example of a near-eye display device including a waveguide display.
- (7) FIG. 5B illustrates an example of a near-eye display device including a waveguide display according to certain embodiments.
- (8) FIG. 6 illustrates an example of an image source assembly in an augmented reality system.
- (9) FIG. 7 illustrates an example of a light beam scanning-based head-mounted display system.
- (10) FIG. 8A illustrates an example of a raster scanning pattern.
- (11) FIG. 8B illustrates an example of displaying an image using a pulsed light beam and a raster

scanning pattern.

(12) FIG. 8C illustrates examples of scanning patterns.

(13) FIG. 8D illustrates an example of displaying an image using a pulsed light beam and a Lissajous scanning pattern.

(14) FIG. 9 illustrates an example of a vector scanning-based head-mounted display system according to certain embodiments.

(15) FIGS. 10A-10D illustrate examples of displaying an image using vector scanning according to certain embodiments.

(16) FIG. 11A illustrates an example of a vector scanning-based head-mounted display system with a variable beam width according to certain embodiments.

(17) FIG. 11B illustrates an example of displaying an image by vector scanning using a light beam having a variable beam width according to certain embodiments.

(18) FIGS. 12A-12C illustrate an example of a method of displaying an image by vector scanning according to certain embodiments.

(19) FIG. 13 illustrates an example of a waveguide-based head-mounted display system according to certain embodiments.

(20) FIG. 14A illustrates an example of a vector scanning-based head-mounted display system capable of non-uniformity correction according to certain embodiments.

(21) FIG. 14B illustrates an example of non-uniformity correction using the vector scanning-based head-mounted display system of FIG. 14A according to certain embodiments.

(22) FIG. 15 includes a flowchart illustrating an example of a method of displaying an image using a vector scanning-based head-mounted display system according to certain embodiments.

(23) FIG. 16 is a simplified block diagram of an electronic system of an example of a near-eye display according to certain embodiments.

(24) In the appended figures, similar components and/or features may have the same reference label. Further, various components of the same type may be distinguished by following the reference label by a dash and a second label that distinguishes among the similar components. If only the first reference label is used in the specification, the description is applicable to any one of the similar components having the same first reference label irrespective of the second reference label.

DETAILED DESCRIPTION

(25) This disclosure relates generally to near-eye display. More specifically, and without limitation, techniques disclosed herein relate to vector scanning-based near-eye display. Various inventive embodiments are described herein, including devices, systems, structures, methods, processes, program code, and the like.

(26) Augmented reality (AR), virtual reality (VR), mixed reality (MR), and other artificial reality applications may use head-mounted display (HMD) systems (which may also be referred to as near-eye display systems) to present images of virtual objects and/or real objects to the user's eyes. A head-mounted display system generally includes an image source (e.g., a display panel) that is near the user's eye and can generate images to be viewed by the user. The head-mounted display system may also include an optical system configured to relay the images generated by the image source to create virtual images that appear to be away from the image source (e.g., further than just a few centimeters away from the user's eyes). The image source of the head-mounted display system may include, for example, a liquid crystal display (LCD), an organic light emitting diode (OLED) display, a micro-OLED display, an inorganic light emitting diode (ILED) display, a micro light emitting diode (micro-LED) display, an active-matrix OLED display (AMOLED), a transparent OLED display (TOLED), and the like. The optical system of the head-mounted display system may include, for example, a lens (e.g., pancake lens) and/or an optical combiner such as a waveguide combiner, a partial reflector combiner, a prism birdbath combiner, a free-space birdbath combiner, and the like.

(27) It is generally desirable that the head-mounted display system has a small size, a low weight, a large field of view, a large eye box, a high power efficiency, a high image brightness, a high resolution, a high refresh rate, a low cost, a long battery life, and a long lifetime. To achieve such performance, it is generally desirable that the image source (e.g., a display panel) of the head-mounted display system has a higher resolution, a large color gamut, a large field of view (FOV), a high brightness, and better image quality, in order to improve the immersive experience of using the head-mounted display system. For a battery-powered head-mounted display system, it may also be desirable that the light source or image source has a higher power efficiency to improve the battery life of the system. Image sources such as LCD displays, OLED displays, micro-OLED displays, ILED displays, QLED displays, micro-LED displays, AMOLED displays, TOLED displays, and the like, may have limited resolution, brightness, field of view, and power efficiency, and/or may have high costs in order to achieve better resolution, brightness, field of view, and power efficiency. For example, to achieve a high resolution, a large number (e.g., millions) of light emitters may be used, and the displays may be relatively large due to finite sizes of the light emitters and may have high power consumption due to the large number of light emitters. Large display optics may need to be used for large displays. Thus, the head-mounted display system may have a large form factor and a high weight.

(28) Light beam scanning-based head-mounted display systems may display images to users by, for example, steering one or more light beams according to a scanning pattern. The light beams can include laser beams that may have high intensities and high spectral purity. The light beams can be steered over a large angular range to provide a large field of view. Therefore, light beam scanning-based head-mounted display systems may be able to achieve the desired resolution, brightness, field of view, power efficiency, and the like, with a relatively low system complexity and cost. In some examples, two-dimensional (2D) scanning may be used to provide a 2D FOV without using a pixelated light source or image source. For example, in some 2D scanning-based head-mounted display systems, a collimated laser beam may be steered toward different directions across the field of view to display an image. Each pixel of the image may be displayed by steering the laser beam towards a respective direction. In this way, pixels of the image may be displayed sequentially (pixel-by-pixel) by scanning the laser beam. Sequential pixel displaying can be performed at fast speeds to achieve sufficiently high frame rates. For instance, to display images with a resolution of 1000×1000 pixels (1,000,000 pixels) at a frame rate of 60 Hz, each pixel must be displayed within about 16.67 ns ($1/60/1,000,000$) on average.

(29) Some 2D scanning systems may use “raster” or “Lissajous” scanning patterns to direct the laser beam toward the respective direction for each pixel. Some 2D scanning systems may employ micro-electro-mechanical system (MEMS) mirrors to steer light beams by rotating the mirrors around one or two axes. The 2D scanning may be achieved using either dual-axis MEMS mirrors or cascaded single-axis MEMS mirrors. A resonant axis, or two resonant axes in the case of Lissajous scanning, may be used to achieve sufficiently high scanning speeds using mechanical rotation resonance in MEMS mirrors. Some other beam steering devices may include, for example, grating light valves, phased arrays, and the like. These 2D scanning systems may be pixelated scanning systems, where the scanning path may need to cover every pixel, whether or not the pixel needs to be on in a specific image frame. Therefore, the laser beam dwell time at each pixel may be constant, whether the displayed image has a low or high fill factor (e.g., the ratio between the number of pixels that are on and the total number of pixels).

(30) In some of these pixelated light beam scanning-based head-mounted display systems, high-speed laser modulation may be used to generate short-pulsed laser beams for respective pixels to project arbitrary content. A bright pixel may be displayed to the user when the laser emits a short laser pulse while the scanner points to a corresponding direction for the pixel. The brightness of each bright pixel may be limited due to the short dwell time and short laser pulse for each pixel (e.g., less than about 20 ns or less than about 10 ns), whether the displayed image has a low or high

fill factor for bright pixels. To achieve a high brightness, the short-pulsed laser beam may need to have high peak and average optical power. In general, higher electrical driving power may be needed to achieve higher optical power (or brightness) in these short-pulsed laser sources. In addition, the optical systems of the light beam scanning-based head-mounted display systems may need to have high damage thresholds in order to handle the high-power short-pulsed laser beams without being damaged. Therefore, the brightness of the displayed image may be limited by, for example, the driving power and/or the damage threshold of the light beam scanning-based head-mounted display system.

(31) In some augmented reality head-mounted display systems, including some light beam scanning-based head-mounted display systems, a waveguide may be used as an optical combiner to, for example, provide optical see through capability and pupil replication/expansion capability (e.g., to provide a large eyepiece). But some waveguide-based head-mounted display systems may have low overall optical efficiencies (e.g., the efficiency on a path from the light source, through an input coupler, waveguide, and output coupler, to the eyepiece), such as about 1% or a few percents, due to, for example, low efficiencies of the input and output couplers. Due to the lower optical efficiencies of waveguide-based head-mounted display systems, the light sources may need to have higher output power in order to achieve the desired brightness in the images displayed to the eyepiece of the waveguide-based head-mounted display systems. To achieve the higher output power, the waveguide-based head-mounted display systems may need to have higher electrical driving power and higher laser damage thresholds as described above.

(32) In many artificial reality applications, such as many augmented reality displays, the images to be displayed may have low fill factors, where the content may be displayed using only a small portion of the available pixels for each image frame. However, the “raster” or “Lissajous” scanning techniques may still need to scan to all pixel directions, and the light beam dwell time for each pixel may be constant and short, because the scanning path is fixed and is content independent. Therefore, to achieve the desired image brightness, the short-pulsed laser beams may need to have high peak power, and thus the head-mounted display systems may need to have high electrical peak power and high laser damage thresholds. In addition, scanning to corresponding directions for all pixel regardless of the content to be displayed may also consume more electrical power for driving the scanning elements (e.g., MEMS mirrors) of the head-mounted display systems.

(33) According to certain embodiments disclosed herein, vector scanning techniques may be used in some head-mounted display systems (e.g., some augmented reality or mixed reality display systems that include waveguide displays) to steer a light beam according to a scan path that may be content dependent, where the light beam may not need to be steered into all directions for displaying most 2D images. In vector scanning, the content to be displayed may be defined using coordinates and lines, rather than pixels in a pixel grid, where the light beam may function as a paintbrush and may be steered by a beam steering device according to a scan path to paint an image. The beam steering device may be controlled to steer light according to an arbitrary content-dependent scan path, rather than according to a fixed scan path as in “raster” scanning or “Lissajous” scanning. The scan path may be determined based on the content to be displayed. To display different content, different vector scan paths may be used. The color, brightness, pulse duration, and the width of the light beam may be controlled during the scanning process to generate the desired display content.

(34) In vector scanning techniques disclosed herein, the scan path may be optimized to reduce the overall length of the scan path. The light beam may not need to be turned on and off for each pixel direction, and can remain on when the light beam is steered along a straight or curved line. The light source can be turned off and the steering can be at a faster speed during transitions between content of the image, thereby reducing power consumption and increasing the dwell time for content to be displayed. Therefore, compared with “raster” scanning or “Lissajous” scanning techniques, vector scanning techniques disclosed herein may display an image using a content-

dependent scan path that may minimize the scan path length and reduce the power used for the scanning. Because the scan path length may be reduced and the light beam may not need to be steered into all directions for displaying a 2D image, the effective dwell time of the light beam at each steering direction can be increased to increase the brightness of each image frame. Therefore, to achieve the desired brightness of the displayed image, the optical output power of the light source can be reduced. For example, to display a low-fill image that includes only one line at a frame rate of 60 Hz and an effective resolution of 1000×1000 pixels, the dwell time at each scanning direction can be about $16.67 \mu\text{s}$ ($1/60/1000$), rather than about 16.67 ns as in raster scanning or “Lissajous” scanning. As such, with the same optical output power of the light beam, the brightness of the content displayed using vector scanning can be about 1000 times of that of the content displayed using raster scanning or “Lissajous” scanning. Alternatively, compared with raster scanning or “Lissajous” scanning, the same or higher brightness can be achieved by vector scanning using a light beam with a much lower power.

(35) Furthermore, in some examples, the scan path can be modified to include curves or loops (rather than sharp turns) in the scan path. Using curves or loops (rather than sharp turns) may reduce the electrical power used for steering the light beam. For example, in MEMS vector scanner, there may be a relationship between the torque needed to actuate (e.g., rotate) the MEMS mirror and the drive power. The scan path may be optimized to maintain sharp corners in the content while reducing the electrical power used for steering the light beam, for example, by adding a loop at a sharp corner on the scan path and turning off the illumination source during the steering according to the loop. Using curves or loops (rather than sharp turns) in the scan path can reduce the torque for actuating the MEMS mirror, and thus may reduce the electrical power for actuating the MEMS mirror. Using curves or loops at sharp corners may also enable the scanner to scan at a higher speed at the sharp corners.

(36) In addition, in some head-mounted display systems disclosed herein, the light beam can be controlled to have different beam widths during the scanning process, such that the lines of an image generated in a single scan path may have different thicknesses. Furthermore, the brightness of the generated image may be controlled during the scanning process, for example, by controlling the scanning speed or the dwell time at each scanning direction, to pre-compensate brightness nonuniformity that may otherwise be caused by a waveguide display, thereby improving the brightness uniformity of the image output from the waveguide display to the eyepiece. The brightness uniformity correction can be implemented independent of the eye pupil position by improving brightness uniformity across the full eyepiece, or can be implemented with eye tracking (e.g., based on the position and/or field of view of the user's eye).

(37) The vector scanning techniques described herein may be used in conjunction with various technologies, such as an artificial reality system. An artificial reality system, such as a head-mounted display (HMD) or heads-up display (HUD) system, generally includes a display configured to present artificial images that depict objects in a virtual environment. The display may present virtual objects or combine images of real objects with virtual objects, as in virtual reality (VR), augmented reality (AR), or mixed reality (MR) applications. For example, in an AR system, a user may view both displayed images of virtual objects (e.g., computer-generated images (CGIs)) and the surrounding environment by, for example, seeing through transparent display glasses or lenses (often referred to as optical see-through) or viewing displayed images of the surrounding environment captured by a camera (often referred to as video see-through). In some AR systems, the artificial images may be presented to users using an LED-based display subsystem.

(38) In the following description, for the purposes of explanation, specific details are set forth in order to provide a thorough understanding of examples of the disclosure. However, it will be apparent that various examples may be practiced without these specific details. For example, devices, systems, structures, assemblies, methods, and other components may be shown as components in block diagram form in order not to obscure the examples in unnecessary detail. In

other instances, well-known devices, processes, systems, structures, and techniques may be shown without necessary detail in order to avoid obscuring the examples. The figures and description are not intended to be restrictive. The terms and expressions that have been employed in this disclosure are used as terms of description and not of limitation, and there is no intention in the use of such terms and expressions of excluding any equivalents of the features shown and described or portions thereof. The word “example” is used herein to mean “serving as an example, instance, or illustration.” Any embodiment or design described herein as “example” is not necessarily to be construed as preferred or advantageous over other embodiments or designs.

(39) FIG. 1 is a simplified block diagram of an example of an artificial reality system environment **100** including a near-eye display **120** in accordance with certain embodiments. Artificial reality system environment **100** shown in FIG. 1 may include near-eye display **120**, an optional external imaging device **150**, and an optional input/output interface **140**, each of which may be coupled to an optional console **110**. While FIG. 1 shows an example of artificial reality system environment **100** including one near-eye display **120**, one external imaging device **150**, and one input/output interface **140**, any number of these components may be included in artificial reality system environment **100**, or any of the components may be omitted. For example, there may be multiple near-eye displays **120** monitored by one or more external imaging devices **150** in communication with console **110**. In some configurations, artificial reality system environment **100** may not include external imaging device **150**, optional input/output interface **140**, and optional console **110**. In alternative configurations, different or additional components may be included in artificial reality system environment **100**.

(40) Near-eye display **120** may be a head-mounted display that presents content to a user. Examples of content presented by near-eye display **120** include one or more of images, videos, audio, or any combination thereof. In some embodiments, audio may be presented via an external device (e.g., speakers and/or headphones) that receives audio information from near-eye display **120**, console **110**, or both, and presents audio data based on the audio information. Near-eye display **120** may include one or more rigid bodies, which may be rigidly or non-rigidly coupled to each other. A rigid coupling between rigid bodies may cause the coupled rigid bodies to act as a single rigid entity. A non-rigid coupling between rigid bodies may allow the rigid bodies to move relative to each other. In various embodiments, near-eye display **120** may be implemented in any suitable form-factor, including a pair of glasses. Some embodiments of near-eye display **120** are further described below with respect to FIGS. 2 and 3. Additionally, in various embodiments, the functionality described herein may be used in a headset that combines images of an environment external to near-eye display **120** and artificial reality content (e.g., computer-generated images). Therefore, near-eye display **120** may augment images of a physical, real-world environment external to near-eye display **120** with generated content (e.g., images, video, sound, etc.) to present an augmented reality to a user.

(41) In various embodiments, near-eye display **120** may include one or more of display electronics **122**, display optics **124**, and an eye-tracking unit **130**. In some embodiments, near-eye display **120** may also include one or more locators **126**, one or more position sensors **128**, and an inertial measurement unit (IMU) **132**. Near-eye display **120** may omit any of eye-tracking unit **130**, locators **126**, position sensors **128**, and IMU **132**, or include additional elements in various embodiments. Additionally, in some embodiments, near-eye display **120** may include elements combining the function of various elements described in conjunction with FIG. 1.

(42) Display electronics **122** may display or facilitate the display of images to the user according to data received from, for example, console **110**. In various embodiments, display electronics **122** may include one or more display panels, such as a liquid crystal display (LCD), an organic light emitting diode (OLED) display, an inorganic light emitting diode (ILED) display, a micro light emitting diode (pLED) display, an active-matrix OLED display (AMOLED), a transparent OLED display (TOLED), or some other display. For example, in one implementation of near-eye display

120, display electronics **122** may include a front TOLED panel, a rear display panel, and an optical component (e.g., an attenuator, polarizer, or diffractive or spectral film) between the front and rear display panels. Display electronics **122** may include pixels to emit light of a predominant color such as red, green, blue, white, or yellow. In some implementations, display electronics **122** may display a three-dimensional (3D) image through stereoscopic effects produced by two-dimensional panels to create a subjective perception of image depth. For example, display electronics **122** may include a left display and a right display positioned in front of a user's left eye and right eye, respectively. The left and right displays may present copies of an image shifted horizontally relative to each other to create a stereoscopic effect (i.e., a perception of image depth by a user viewing the image).

(43) In certain embodiments, display optics **124** may display image content optically (e.g., using optical waveguides and couplers) or magnify image light received from display electronics **122**, correct optical errors associated with the image light, and present the corrected image light to a user of near-eye display **120**. In various embodiments, display optics **124** may include one or more optical elements, such as, for example, a substrate, optical waveguides, an aperture, a Fresnel lens, a convex lens, a concave lens, a filter, input/output couplers, or any other suitable optical elements that may affect image light emitted from display electronics **122**. Display optics **124** may include a combination of different optical elements as well as mechanical couplings to maintain relative spacing and orientation of the optical elements in the combination. One or more optical elements in display optics **124** may have an optical coating, such as an antireflective coating, a reflective coating, a filtering coating, or a combination of different optical coatings.

(44) Locators **126** may be objects located in specific positions on near-eye display **120** relative to one another and relative to a reference point on near-eye display **120**. In some implementations, console **110** may identify locators **126** in images captured by external imaging device **150** to determine the artificial reality headset's position, orientation, or both. A locator **126** may be an LED, a corner cube reflector, a reflective marker, a type of light source that contrasts with an environment in which near-eye display **120** operates, or any combination thereof. In embodiments where locators **126** are active components (e.g., LEDs or other types of light emitting devices).

(45) External imaging device **150** may include one or more cameras, one or more video cameras, any other device capable of capturing images including one or more of locators **126**, or any combination thereof. Additionally, external imaging device **150** may include one or more filters (e.g., to increase signal to noise ratio). External imaging device **150** may be configured to detect light emitted or reflected from locators **126** in a field of view of external imaging device **150**. In embodiments where locators **126** include passive elements (e.g., retroreflectors), external imaging device **150** may include a light source that illuminates some or all of locators **126**, which may retro-reflect the light to the light source in external imaging device **150**. Slow calibration data may be communicated from external imaging device **150** to console **110**, and external imaging device **150** may receive one or more calibration parameters from console **110** to adjust one or more imaging parameters (e.g., focal length, focus, frame rate, sensor temperature, shutter speed, aperture, etc.).

(46) Position sensors **128** may generate one or more measurement signals in response to motion of near-eye display **120**. Examples of position sensors **128** may include accelerometers, gyroscopes, magnetometers, other motion-detecting or error-correcting sensors, or any combination thereof. For example, in some embodiments, position sensors **128** may include multiple accelerometers to measure translational motion (e.g., forward/back, up/down, or left/right) and multiple gyroscopes to measure rotational motion (e.g., pitch, yaw, or roll). In some embodiments, various position sensors may be oriented orthogonally to each other.

(47) IMU **132** may be an electronic device that generates fast calibration data based on measurement signals received from one or more of position sensors **128**. Position sensors **128** may be located external to IMU **132**, internal to IMU **132**, or any combination thereof. Based on the one

or more measurement signals from one or more position sensors **128**, IMU **132** may generate fast calibration data indicating an estimated position of near-eye display **120** relative to an initial position of near-eye display **120**.

(48) Eye-tracking unit **130** may include one or more eye-tracking systems. Eye tracking may refer to determining an eye's position, including orientation and location of the eye, relative to near-eye display **120**. An eye-tracking system may include an imaging system to image one or more eyes and may optionally include a light emitter, which may generate light that is directed to an eye such that light reflected by the eye may be captured by the imaging system. Near-eye display **120** may use the orientation of the eye to, e.g., determine an inter-pupillary distance (IPD) of the user, determine gaze direction, introduce depth cues (e.g., blur image outside of the user's main line of sight), collect heuristics on the user interaction in the VR media (e.g., time spent on any particular subject, object, or frame as a function of exposed stimuli), some other functions that are based in part on the orientation of at least one of the user's eyes, or any combination thereof.

(49) Input/output interface **140** may be a device that allows a user to send action requests to console **110**. An action request may be a request to perform a particular action. For example, an action request may be to start or to end an application or to perform a particular action within the application. Input/output interface **140** may include one or more input devices. Example input devices may include a keyboard, a mouse, a game controller, a glove, a button, a touch screen, or any other suitable device for receiving action requests and communicating the received action requests to console **110**. An action request received by the input/output interface **140** may be communicated to console **110**, which may perform an action corresponding to the requested action. In some embodiments, input/output interface **140** may provide haptic feedback to the user in accordance with instructions received from console **110**. In some embodiments, external imaging device **150** may be used to track input/output interface **140**, such as tracking the location or position of a controller (which may include, for example, an IR light source) or a hand of the user to determine the motion of the user. In some embodiments, near-eye display **120** may include one or more imaging devices to track input/output interface **140**, such as tracking the location or position of a controller or a hand of the user to determine the motion of the user.

(50) Console **110** may provide content to near-eye display **120** for presentation to the user in accordance with information received from one or more of external imaging device **150**, near-eye display **120**, and input/output interface **140**. In the example shown in FIG. **1**, console **110** may include an application store **112**, a headset tracking module **114**, an artificial reality engine **116**, and an eye-tracking module **118**. Some embodiments of console **110** may include different or additional modules than those described in conjunction with FIG. **1**. Functions further described below may be distributed among components of console **110** in a different manner than is described here.

(51) In some embodiments, console **110** may include a processor and a non-transitory computer-readable storage medium storing instructions executable by the processor. The processor may include multiple processing units executing instructions in parallel. The non-transitory computer-readable storage medium may be any memory, such as a hard disk drive, a removable memory, or a solid-state drive (e.g., flash memory or dynamic random access memory (DRAM)). In various embodiments, the modules of console **110** described in conjunction with FIG. **1** may be encoded as instructions in the non-transitory computer-readable storage medium that, when executed by the processor, cause the processor to perform the functions further described below.

(52) Application store **112** may store one or more applications for execution by console **110**. An application may include a group of instructions that, when executed by a processor, generates content for presentation to the user. Content generated by an application may be in response to inputs received from the user via movement of the user's eyes or inputs received from the input/output interface **140**. Examples of the applications may include gaming applications, conferencing applications, video playback application, or other suitable applications.

(53) Headset tracking module **114** may track movements of near-eye display **120** using slow

calibration information from external imaging device **150**. For example, headset tracking module **114** may determine positions of a reference point of near-eye display **120** using observed locators from the slow calibration information and a model of near-eye display **120**. Headset tracking module **114** may also determine positions of a reference point of near-eye display **120** using position information from the fast calibration information. Additionally, in some embodiments, headset tracking module **114** may use portions of the fast calibration information, the slow calibration information, or any combination thereof, to predict a future location of near-eye display **120**. Headset tracking module **114** may provide the estimated or predicted future position of near-eye display **120** to artificial reality engine **116**.

(54) Artificial reality engine **116** may execute applications within artificial reality system environment **100** and receive position information of near-eye display **120**, acceleration information of near-eye display **120**, velocity information of near-eye display **120**, predicted future positions of near-eye display **120**, or any combination thereof from headset tracking module **114**. Artificial reality engine **116** may also receive estimated eye position and orientation information from eye-tracking module **118**. Based on the received information, artificial reality engine **116** may determine content to provide to near-eye display **120** for presentation to the user. Artificial reality engine **116** may perform an action within an application executing on console **110** in response to an action request received from input/output interface **140**, and provide feedback to the user indicating that the action has been performed. The feedback may be visual or audible feedback via near-eye display **120** or haptic feedback via input/output interface **140**.

(55) Eye-tracking module **118** may receive eye-tracking data from eye-tracking unit **130** and determine the position of the user's eye based on the eye tracking data. The position of the eye may include an eye's orientation, location, or both relative to near-eye display **120** or any element thereof. Because the eye's axes of rotation change as a function of the eye's location in its socket, determining the eye's location in its socket may allow eye-tracking module **118** to determine the eye's orientation more accurately.

(56) FIG. 2 is a perspective view of an example of a near-eye display in the form of an HMD device **200** for implementing some of the examples disclosed herein. HMD device **200** may be a part of, e.g., a VR system, an AR system, an MR system, or any combination thereof. HMD device **200** may include a body **220** and a head strap **230**. FIG. 2 shows a bottom side **223**, a front side **225**, and a left side **227** of body **220** in the perspective view. Head strap **230** may have an adjustable or extendible length. There may be a sufficient space between body **220** and head strap **230** of HMD device **200** for allowing a user to mount HMD device **200** onto the user's head. In various embodiments, HMD device **200** may include additional, fewer, or different components. For example, in some embodiments, HMD device **200** may include eyeglass temples and temple tips as shown in, for example, FIG. 3 below, rather than head strap **230**.

(57) HMD device **200** may present to a user media including virtual and/or augmented views of a physical, real-world environment with computer-generated elements. Examples of the media presented by HMD device **200** may include images (e.g., two-dimensional (2D) or three-dimensional (3D) images), videos (e.g., 2D or 3D videos), audio, or any combination thereof. The images and videos may be presented to each eye of the user by one or more display assemblies (not shown in FIG. 2) enclosed in body **220** of HMD device **200**. In various embodiments, the one or more display assemblies may include a single electronic display panel or multiple electronic display panels (e.g., one display panel for each eye of the user). Examples of the electronic display panel(s) may include, for example, an LCD, an OLED display, an ILED display, a pLED display, an AMOLED, a TOLED, some other display, or any combination thereof. HMD device **200** may include two eye box regions.

(58) In some implementations, HMD device **200** may include various sensors (not shown), such as depth sensors, motion sensors, position sensors, and eye tracking sensors. Some of these sensors may use a structured light pattern for sensing. In some implementations, HMD device **200** may

include an input/output interface for communicating with a console. In some implementations, HMD device **200** may include a virtual reality engine (not shown) that can execute applications within HMD device **200** and receive depth information, position information, acceleration information, velocity information, predicted future positions, or any combination thereof of HMD device **200** from the various sensors. In some implementations, the information received by the virtual reality engine may be used for producing a signal (e.g., display instructions) to the one or more display assemblies. In some implementations, HMD device **200** may include locators (not shown, such as locators **126**) located in fixed positions on body **220** relative to one another and relative to a reference point. Each of the locators may emit light that is detectable by an external imaging device.

(59) FIG. **3** is a perspective view of an example of a near-eye display **300** in the form of a pair of glasses for implementing some of the examples disclosed herein. Near-eye display **300** may be a specific implementation of near-eye display **120** of FIG. **1**, and may be configured to operate as a virtual reality display, an augmented reality display, and/or a mixed reality display. Near-eye display **300** may include a frame **305** and a display **310**. Display **310** may be configured to present content to a user. In some embodiments, display **310** may include display electronics and/or display optics. For example, as described above with respect to near-eye display **120** of FIG. **1**, display **310** may include an LCD display panel, an LED display panel, or an optical display panel (e.g., a waveguide display assembly).

(60) Near-eye display **300** may further include various sensors **350a**, **350b**, **350c**, **350d**, and **350e** on or within frame **305**. In some embodiments, sensors **350a-350e** may include one or more depth sensors, motion sensors, position sensors, inertial sensors, or ambient light sensors. In some embodiments, sensors **350a-350e** may include one or more image sensors configured to generate image data representing different fields of views in different directions. In some embodiments, sensors **350a-350e** may be used as input devices to control or influence the displayed content of near-eye display **300**, and/or to provide an interactive VR/AR/MR experience to a user of near-eye display **300**. In some embodiments, sensors **350a-350e** may also be used for stereoscopic imaging.

(61) In some embodiments, near-eye display **300** may further include one or more illuminators **330** to project light into the physical environment. The projected light may be associated with different frequency bands (e.g., visible light, infra-red light, ultra-violet light, etc.), and may serve various purposes. For example, illuminator(s) **330** may project light in a dark environment (or in an environment with low intensity of infra-red light, ultra-violet light, etc.) to assist sensors **350a-350e** in capturing images of different objects within the dark environment. In some embodiments, illuminator(s) **330** may be used to project certain light patterns onto the objects within the environment. In some embodiments, illuminator(s) **330** may be used as locators, such as locators **126** described above with respect to FIG. **1**.

(62) In some embodiments, near-eye display **300** may also include a high-resolution camera **340**. Camera **340** may capture images of the physical environment in the field of view. The captured images may be processed, for example, by a virtual reality engine (e.g., artificial reality engine **116** of FIG. **1**) to add virtual objects to the captured images or modify physical objects in the captured images, and the processed images may be displayed to the user by display **310** for AR or MR applications.

(63) FIG. **4** illustrates an example of an optical see-through augmented reality system **400** including a waveguide display according to certain embodiments. Augmented reality system **400** may include a projector **410** and a combiner **415**. Projector **410** may include a light source or image source **412** and projector optics **414**. In some embodiments, light source or image source **412** may include one or more micro-LED devices described above. In some embodiments, image source **412** may include a plurality of pixels that displays virtual objects, such as an LCD display panel or an LED display panel. In some embodiments, image source **412** may include a light source that generates coherent or partially coherent light. For example, image source **412** may include a laser

diode, a vertical cavity surface emitting laser, an LED, and/or a micro-LED described above. In some embodiments, image source **412** may include a plurality of light sources (e.g., an array of micro-LEDs described above), each emitting a monochromatic image light corresponding to a primary color (e.g., red, green, or blue). In some embodiments, image source **412** may include three two-dimensional arrays of micro-LEDs, where each two-dimensional array of micro-LEDs may include micro-LEDs configured to emit light of a primary color (e.g., red, green, or blue). In some embodiments, image source **412** may include an optical pattern generator, such as a spatial light modulator. Projector optics **414** may include one or more optical components that can condition the light from image source **412**, such as expanding, collimating, scanning, or projecting light from image source **412** to combiner **415**. The one or more optical components may include, for example, one or more lenses, liquid lenses, mirrors, apertures, and/or gratings. For example, in some embodiments, image source **412** may include one or more one-dimensional arrays or elongated two-dimensional arrays of micro-LEDs, and projector optics **414** may include one or more one-dimensional scanners (e.g., micro-mirrors or prisms) configured to scan the one-dimensional arrays or elongated two-dimensional arrays of micro-LEDs to generate image frames. In some embodiments, projector optics **414** may include a liquid lens (e.g., a liquid crystal lens) with a plurality of electrodes that allows scanning of the light from image source **412**.

(64) Combiner **415** may include an input coupler **430** for coupling light from projector **410** into a substrate **420** of combiner **415**. Input coupler **430** may include a volume holographic grating, a diffractive optical element (DOE) (e.g., a surface-relief grating), a slanted surface of substrate **420**, or a refractive coupler (e.g., a wedge or a prism). For example, input coupler **430** may include a reflective volume Bragg grating or a transmissive volume Bragg grating. Input coupler **430** may have a coupling efficiency of greater than 30%, 50%, 75%, 90%, or higher for visible light. Light coupled into substrate **420** may propagate within substrate **420** through, for example, total internal reflection (TIR). Substrate **420** may be in the form of a lens of a pair of eyeglasses. Substrate **420** may have a flat or a curved surface, and may include one or more types of dielectric materials, such as glass, quartz, plastic, polymer, poly(methyl methacrylate) (PMMA), crystal, or ceramic. A thickness of the substrate may range from, for example, less than about 1 mm to about 10 mm or more. Substrate **420** may be transparent to visible light.

(65) Substrate **420** may include or may be coupled to a plurality of output couplers **440**, each configured to extract at least a portion of the light guided by and propagating within substrate **420** from substrate **420**, and direct extracted light **460** to an eyebox **495** where an eye **490** of the user of augmented reality system **400** may be located when augmented reality system **400** is in use. The plurality of output couplers **440** may replicate the exit pupil to increase the size of eyebox **495** such that the displayed image is visible in a larger area. As input coupler **430**, output couplers **440** may include grating couplers (e.g., volume holographic gratings or surface-relief gratings), other diffraction optical elements (DOEs), prisms, etc. For example, output couplers **440** may include reflective volume Bragg gratings or transmissive volume Bragg gratings. Output couplers **440** may have different coupling (e.g., diffraction) efficiencies at different locations. Substrate **420** may also allow light **450** from the environment in front of combiner **415** to pass through with little or no loss. Output couplers **440** may also allow light **450** to pass through with little loss. For example, in some implementations, output couplers **440** may have a very low diffraction efficiency for light **450** such that light **450** may be refracted or otherwise pass through output couplers **440** with little loss, and thus may have a higher intensity than extracted light **460**. In some implementations, output couplers **440** may have a high diffraction efficiency for light **450** and may diffract light **450** in certain desired directions (i.e., diffraction angles) with little loss. As a result, the user may be able to view combined images of the environment in front of combiner **415** and images of virtual objects projected by projector **410**.

(66) FIG. 5A illustrates an example of a near-eye display (NED) device **500** including a waveguide display **530** according to certain embodiments. NED device **500** may be an example of near-eye

display **120**, augmented reality system **400**, or another type of display device. NED device **500** may include a light source **510**, projection optics **520**, and waveguide display **530**. Light source **510** may include multiple panels of light emitters for different colors, such as a panel of red light emitters **512**, a panel of green light emitters **514**, and a panel of blue light emitters **516**. The red light emitters **512** are organized into an array; the green light emitters **514** are organized into an array; and the blue light emitters **516** are organized into an array. The dimensions and pitches of light emitters in light source **510** may be small. For example, each light emitter may have a diameter less than $2\text{ }\mu\text{m}$ (e.g., about $1.2\text{ }\mu\text{m}$) and the pitch may be less than $2\text{ }\mu\text{m}$ (e.g., about $1.5\text{ }\mu\text{m}$). As such, the number of light emitters in each red light emitters **512**, green light emitters **514**, and blue light emitters **516** can be equal to or greater than the number of pixels in a display image, such as 960×720 , 1280×720 , 1440×1080 , 1920×1080 , 2160×1080 , 2560×1080 , or more pixels. Thus, a display image may be generated simultaneously by light source **510**. A scanning element may not be used in NED device **500**.

(67) Before reaching waveguide display **530**, the light emitted by light source **510** may be conditioned by projection optics **520**, which may include a lens array. Projection optics **520** may collimate or focus the light emitted by light source **510** to waveguide display **530**, which may include a coupler **532** for coupling the light emitted by light source **510** into waveguide display **530**. The light coupled into waveguide display **530** may propagate within waveguide display **530** through, for example, total internal reflection as described above with respect to FIG. **4**. Coupler **532** may also couple portions of the light propagating within waveguide display **530** out of waveguide display **530** and towards user's eye **590**.

(68) FIG. **5B** illustrates an example of a near-eye display (NED) device **550** including a waveguide display **580** according to certain embodiments. In some embodiments, NED device **550** may use a scanning mirror **570** to project light from a light source **540** to an image field where a user's eye **590** may be located. NED device **550** may be an example of near-eye display **120**, augmented reality system **400**, or another type of display device. Light source **540** may include one or more rows or one or more columns of light emitters of different colors, such as multiple rows of red light emitters **542**, multiple rows of green light emitters **544**, and multiple rows of blue light emitters **546**. For example, red light emitters **542**, green light emitters **544**, and blue light emitters **546** may each include N rows, each row including, for example, 2560 light emitters (pixels). The red light emitters **542** are organized into an array; the green light emitters **544** are organized into an array; and the blue light emitters **546** are organized into an array. In some embodiments, light source **540** may include a single line of light emitters for each color. In some embodiments, light source **540** may include multiple columns of light emitters for each of red, green, and blue colors, where each column may include, for example, 1080 light emitters. In some embodiments, the dimensions and/or pitches of the light emitters in light source **540** may be relatively large (e.g., about $3\text{-}5\text{ }\mu\text{m}$) and thus light source **540** may not include sufficient light emitters for simultaneously generating a full display image. For example, the number of light emitters for a single color may be fewer than the number of pixels (e.g., 2560×1080 pixels) in a display image. The light emitted by light source **540** may be a set of collimated or diverging beams of light.

(69) Before reaching scanning mirror **570**, the light emitted by light source **540** may be conditioned by various optical devices, such as collimating lenses or a freeform optical element **560**. Freeform optical element **560** may include, for example, a multi-facet prism or another light folding element that may direct the light emitted by light source **540** towards scanning mirror **570**, such as changing the propagation direction of the light emitted by light source **540** by, for example, about 90° or larger. In some embodiments, freeform optical element **560** may be rotatable to scan the light. Scanning mirror **570** and/or freeform optical element **560** may reflect and project the light emitted by light source **540** to waveguide display **580**, which may include a coupler **582** for coupling the light emitted by light source **540** into waveguide display **580**. The light coupled into waveguide display **580** may propagate within waveguide display **580** through, for example, total internal

reflection as described above with respect to FIG. 4. Coupler 582 may also couple portions of the light propagating within waveguide display 580 out of waveguide display 580 and towards user's eye 590.

(70) Scanning mirror 570 may include a microelectromechanical system (MEMS) mirror or any other suitable mirrors. Scanning mirror 570 may rotate to scan in one or two dimensions. As scanning mirror 570 rotates, the light emitted by light source 540 may be directed to a different area of waveguide display 580 such that a full display image may be projected onto waveguide display 580 and directed to user's eye 590 by waveguide display 580 in each scanning cycle. For example, in embodiments where light source 540 includes light emitters for all pixels in one or more rows or columns, scanning mirror 570 may be rotated in the column or row direction (e.g., x or y direction) to scan an image. In embodiments where light source 540 includes light emitters for some but not all pixels in one or more rows or columns, scanning mirror 570 may be rotated in both the row and column directions (e.g., both x and y directions) to project a display image (e.g., using a raster-type scanning pattern).

(71) NED device 550 may operate in predefined display periods. A display period (e.g., display cycle) may refer to a duration of time in which a full image is scanned or projected. For example, a display period may be a reciprocal of the desired frame rate. In NED device 550 that includes scanning mirror 570, the display period may also be referred to as a scanning period or scanning cycle. The light generation by light source 540 may be synchronized with the rotation of scanning mirror 570. For example, each scanning cycle may include multiple scanning steps, where light source 540 may generate a different light pattern in each respective scanning step.

(72) In each scanning cycle, as scanning mirror 570 rotates, a display image may be projected onto waveguide display 580 and user's eye 590. The actual color value and light intensity (e.g., brightness) of a given pixel location of the display image may be an average of the light beams of the three colors (e.g., red, green, and blue) illuminating the pixel location during the scanning period. After completing a scanning period, scanning mirror 570 may revert back to the initial position to project light for the first few rows of the next display image or may rotate in a reverse direction or scan pattern to project light for the next display image, where a new set of driving signals may be fed to light source 540. The same process may be repeated as scanning mirror 570 rotates in each scanning cycle. As such, different images may be projected to user's eye 590 in different scanning cycles.

(73) FIG. 6 illustrates an example of an image source assembly 610 in a head-mounted display system 600 according to certain embodiments. Image source assembly 610 may include, for example, a display panel 640 that may generate display images to be projected to the user's eyes, and a projector 650 that may project the display images generated by display panel 640 to a waveguide display as described above with respect to FIGS. 4-5B. Display panel 640 may include a light source 642 and a drive circuit 644 for light source 642. Light source 642 may include, for example, light source 510 or 540. Projector 650 may include, for example, freeform optical element 560, scanning mirror 570, and/or projection optics 520 described above. Head-mounted display system 600 may also include a controller 620 that synchronously controls light source 642 and projector 650 (e.g., scanning mirror 570). Image source assembly 610 may generate and output an image light to a waveguide display (not shown in FIG. 6), such as waveguide display 530 or 580. As described above, the waveguide display may receive the image light at one or more input-coupling elements, and guide the received image light to one or more output-coupling elements. The input and output coupling elements may include, for example, a diffraction grating, a holographic grating, a prism, or any combination thereof. The input-coupling element may be chosen such that total internal reflection occurs with the waveguide display. The output-coupling element may couple portions of the total internally reflected image light out of the waveguide display.

(74) As described above, light source 642 may include a plurality of light emitters arranged in an

array or a matrix. Each light emitter may emit monochromatic light, such as red light, blue light, green light, infra-red light, and the like. While RGB colors are often discussed in this disclosure, embodiments described herein are not limited to using red, green, and blue as primary colors. Other colors can also be used as the primary colors of head-mounted display system **600**. In some embodiments, a display panel in accordance with an embodiment may use more than three primary colors. Each pixel in light source **642** may include three subpixels that include a red micro-LED, a green micro-LED, and a blue micro-LED. A semiconductor LED generally includes an active light emitting layer within multiple layers of semiconductor materials. The multiple layers of semiconductor materials may include different compound materials or a same base material with different dopants and/or different doping densities. For example, the multiple layers of semiconductor materials may include an n-type material layer, an active region that may include hetero-structures (e.g., one or more quantum wells), and a p-type material layer. The multiple layers of semiconductor materials may be grown on a surface of a substrate having a certain orientation.

(75) Controller **620** may control the image rendering operations of image source assembly **610**, such as the operations of light source **642** and/or projector **650**. For example, controller **620** may determine instructions for image source assembly **610** to render one or more display images. The instructions may include display instructions and scanning instructions. In some embodiments, the display instructions may include an image file (e.g., a bitmap file). The display instructions may be received from, for example, a console, such as console **110** described above with respect to FIG. **1**. The scanning instructions may be used by image source assembly **610** to generate image light. The scanning instructions may specify, for example, a type of a source of image light (e.g., monochromatic or polychromatic), a scanning rate, an orientation of a scanning apparatus, one or more illumination parameters, or any combination thereof. Controller **620** may include a combination of hardware, software, and/or firmware not shown here so as not to obscure other aspects of the present disclosure.

(76) In some embodiments, controller **620** may be a graphics processing unit (GPU) of a display device. In other embodiments, controller **620** may be other kinds of processors. The operations performed by controller **620** may include taking content for display and dividing the content into discrete sections. Controller **620** may provide to light source **642** scanning instructions that include an address corresponding to an individual source element of light source **642** and/or an electrical bias applied to the individual source element. Controller **620** may instruct light source **642** to sequentially present the discrete sections using light emitters corresponding to one or more rows of pixels in an image ultimately displayed to the user. Controller **620** may also instruct projector **650** to perform different adjustments of the light. For example, controller **620** may control projector **650** to scan the discrete sections to different areas of a coupling element of the waveguide display (e.g., waveguide display **580**) as described above with respect to FIG. **5B**. As such, at the exit pupil of the waveguide display, each discrete portion is presented in a different respective location. While each discrete section is presented at a different respective time, the presentation and scanning of the discrete sections occur fast enough such that a user's eye may integrate the different sections into a single image or series of images.

(77) Image processor **630** may be a general-purpose processor and/or one or more application-specific circuits that are dedicated to performing the features described herein. In one embodiment, a general-purpose processor may be coupled to a memory to execute software instructions that cause the processor to perform certain processes described herein. In another embodiment, image processor **630** may be one or more circuits that are dedicated to performing certain features. While image processor **630** in FIG. **6** is shown as a stand-alone unit that is separate from controller **620** and drive circuit **644**, image processor **630** may be a sub-unit of controller **620** or drive circuit **644** in other embodiments. In other words, in those embodiments, controller **620** or drive circuit **644** may perform various image processing functions of image processor **630**. Image processor **630**

may also be referred to as an image processing circuit.

(78) In the example shown in FIG. 6, light source **642** may be driven by drive circuit **644**, based on data or instructions (e.g., display and scanning instructions) sent from controller **620** or image processor **630**. In one embodiment, drive circuit **644** may include a circuit panel that connects to and mechanically holds various light emitters of light source **642**. Light source **642** may emit light in accordance with one or more illumination parameters that are set by the controller **620** and potentially adjusted by image processor **630** and drive circuit **644**. An illumination parameter may be used by light source **642** to generate light. An illumination parameter may include, for example, source wavelength, pulse rate, pulse amplitude, beam type (continuous or pulsed), other parameter(s) that may affect the emitted light, or any combination thereof. In some embodiments, the source light generated by light source **642** may include multiple beams of red light, green light, and blue light, or any combination thereof.

(79) Projector **650** may perform a set of optical functions, such as focusing, combining, conditioning, or scanning the image light generated by light source **642**. In some embodiments, projector **650** may include a combining assembly, a light conditioning assembly, or a scanning mirror assembly. Projector **650** may include one or more optical components that optically adjust and potentially re-direct the light from light source **642**. One example of the adjustment of light may include conditioning the light, such as expanding, collimating, correcting for one or more optical errors (e.g., field curvature, chromatic aberration, etc.), some other adjustments of the light, or any combination thereof. The optical components of projector **650** may include, for example, lenses, mirrors, apertures, gratings, or any combination thereof.

(80) Projector **650** may redirect image light via its one or more reflective and/or refractive portions so that the image light is projected at certain orientations toward the waveguide display. The location where the image light is redirected toward the waveguide display may depend on specific orientations of the one or more reflective and/or refractive portions. In some embodiments, projector **650** includes a single scanning mirror that scans in at least two dimensions. In other embodiments, projector **650** may include a plurality of scanning mirrors that each scan in directions orthogonal to each other. Projector **650** may perform a raster scan (horizontally or vertically), a bi-resonant scan, or any combination thereof. In some embodiments, projector **650** may perform a controlled vibration along the horizontal and/or vertical directions with a specific frequency of oscillation to scan along two dimensions and generate a two-dimensional projected image of the media presented to user's eyes. In other embodiments, projector **650** may include a lens or prism that may serve similar or the same function as one or more scanning mirrors. In some embodiments, image source assembly **610** may not include a projector, where the light emitted by light source **642** may be directly incident on the waveguide display.

(81) As described above, light sources or image sources such as LCD displays, OLED displays, micro-OLED displays, ILED displays, QLED displays, micro-LED displays, AMOLED displays, TOLED displays, and the like, may have limited resolution, brightness, field of view, and power efficiency, and/or may have high costs in order to achieve better resolution, brightness, field of view, and power efficiency. To achieve a high resolution, a large number (e.g., a few millions) of light emitters (pixels) may be used, and thus the displays may be relatively large due to finite sizes of the light emitters and may have high power consumption due to the large number of light emitters. Large display optics may also need to be used for large displays. Thus, the head-mounted display system may have a large form factor and a high weight.

(82) Some head-mounted display systems may display images to users by, for example, scanning one or more collimated light beams according to a scanning pattern. In one example, a single light beam (e.g., a laser beam) that may have a high intensity and high spectral purity may be steered to display images to users. Each pixel of the image may be displayed by steering the light beam towards a respective direction. Pixels of the image may be displayed sequentially (pixel-by-pixel) by sequentially steering the light beam towards different directions. The light beams can be steered

at a small angular step over a large angular range to provide a large field of view and high resolution. Therefore, light beam scanning-based head-mounted display systems may be able to achieve the desired resolution (e.g., an angular resolution about 2 arc-min or better), brightness, field of view, power efficiency, and the like, with a relatively low system complexity and cost. In some augmented reality head-mounted display systems, in order to provide optical see-through capabilities and pupil replication capabilities, a waveguide may be used as an optical combiner as described above with respect to, for example, FIGS. 4-5B, where the light beam may be steered to different directions toward the waveguide. The steered light beam may be incident on the waveguide at different incidence angles, and may be coupled into the waveguide at different angles by one or more input couplers to propagate within the waveguide at different directions, for example, through total internal reflection. The light beam propagating within the waveguide at different angles may be coupled out of the waveguide at different directions towards an eyepiece at one or more locations of the waveguide by one or more output couplers, thereby sequentially forming an image in a large eyepiece.

(83) FIG. 7 illustrates an example of a light beam scanning-based head-mounted display system 700 according to certain embodiments. In the illustrated example, head-mounted display system 700 may include a light source 710, a scanner 720, a waveguide display 730, and one or more controllers (not shown) that may control light source 710 and scanner 720. In some examples, light source 710 may include a laser and collimation optics. In some examples, the laser may emit continuous-wave (CW) light. In some examples, the laser or the CW output of the laser may be modulated such that light source 710 may emit a pulsed laser beam that may include a series of short laser pulses. In some examples, the modulation may include modulating the amplitude of the laser beam, turning on and off the output of the laser, generating a series of laser pulses at a certain repetition rate, and the like, or a combination thereof. The laser beam may be collimated by the collimation optics, and may then be steered by scanner 720 according to a scan pattern to generate a two-dimensional image. Scanner 720 may include, for example, a 2D scanning MEMS mirror or two cascaded 1D scanning MEMS mirrors. The modulation of light source 710 and the scanning of scanner 720 may be synchronized so that light of desired brightnesses may be directed to desired directions to display the image. For example, light source 710 and scanner 720 may be controlled by a same central controller, or may be controlled by two controllers that may be synchronized. As illustrated, the laser beam emitted by light source 710 may be directed to one direction at a first time instant and incident on waveguide display 730 at a first incidence angle, and may be directed to a second direction at a second time instant and incident on waveguide display 730 at a second incidence angle.

(84) As described above with respect to, for example, FIGS. 4-5B, waveguide display 730 may include a substrate 732 that may be transparent to visible light, such as a glass substrate, a polycarbonate substrate, a SiC substrate, a plastic substrate, and the like. In some examples, substrate 732 may include two or more sub-layers bonded together. One or more input couplers 734 and one or more output couplers 736 may be formed at the surfaces of substrate 732 or may be embedded in substrate 732 (e.g., between sub-layers). Input couplers 734 and output couplers 736 may include, for example, holographic grating couplers, surface-relief grating couplers, polarization volume holograms (PVHs), and the like. Laser beams incident on input couplers 734 from different directions may be coupled into substrate 732 by input couplers 734, may propagate within substrate 732 at different directions, and may be coupled out of substrate 732 at multiple locations by output couplers 736 to propagate in different directions towards an eyepiece of head-mounted display system 700.

(85) In some examples, scanner 720 may steer the laser beam according to a “raster” scanning or “Lissajous” scanning pattern. Each pixel of the image may be displayed by steering the laser beam towards a respective direction. For pixels that need to be on (bright pixels) in an image, the laser beam may be on when it is steered to the corresponding direction. In some examples, the laser

beam may be turned on or off by modulating the laser source or the output of the laser source, to generate laser pulses with desired durations, where the brightness of each bright pixel may depend on the duration of the laser pulse for the pixel. In some examples, the brightness of each bright pixel may depend on the dwell time of the laser beam for the pixel. In this way, pixels of the image may be displayed sequentially pixel by pixel by steering the laser beam towards different directions. The sequential pixel displaying can be performed at fast speeds to achieve sufficiently high frame rates. For instance, to display images with a resolution of 1000×1000 pixels (1,000,000 pixels) at a frame rate of 60 Hz, each pixel must be displayed within about 16.67 ns ($1/60/1,000,000$) on average.

(86) FIG. 8A illustrates an example of a raster scanning pattern **800**. In the illustrated example, scanning pattern **800** may include a plurality of scan lines **810** and transition lines **820** and **830**. Each scan line **810** may cover a row of pixels, and the plurality of scan lines **810** may cover all pixels in the field of view. The light beam may be at least periodically turned on when it is steered along a scan line **810**. After being steered along each scan line **810**, the light beam may be turned off while the scanner transitions along a transition line **820** from the end of the scan line to the beginning of the next scan line. In some examples, there may be pixels on transition lines **820**, and the laser beam may be at least periodically turned on when the laser beam is steered along transition lines **820**. After the laser beam has been steered along the last scan line **810**, the laser beam may be turned off while the scanner transitions along a transition line **830** from the end of the last scan line **810** to the beginning of the first scan line **810**.

(87) FIG. 8B illustrates an example of displaying an image using a pulsed light beam and a raster scanning pattern. The raster scanning pattern may be similar to or different from the raster scanning pattern shown in FIG. 8A, and may include scan lines **840** that may be traversed from opposite directions as shown by the arrows in FIG. 8B. In the example illustrated in FIG. 8B, a letter “E” may need to be displayed. The laser beam may be steered along scan line **840**, and may include short laser pulses. For pixels that do not need to be on (dark pixels), the laser beam may be turned off when it is steered to directions corresponding to these dark pixels. For each pixel **850** that needs to be on (bright pixel), the laser beam may be turned on for a short period of time (thereby emitting a short laser pulse) when the laser beam is steered to the corresponding direction. As illustrated, the laser beam may need to be steered along a pre-determined path to cover all pixels regardless of the content to be displayed, for example, even if only a few pixels need to be bright. Therefore, the maximum laser beam dwell time (e.g., the duration of a laser pulse) for each pixel may be fixed for a certain image frame rate and image resolution, even if the image to be displayed has a low fill factor (e.g., only a small fraction of all pixels needs to be bright).

(88) FIG. 8C illustrates some other examples of scanning patterns. These examples of scanning patterns S1-S8 may be used to display rectangular image frames, and may be content-independent. Even if the image to be displayed has a low fill factor, the scanning pattern may need to be fully traversed. Therefore, the laser beam dwell time at each pixel may be constant, whether the displayed image has a low or high fill factor for bright pixels. Some of these scanning pattern may include no transition lines or may include short transition lines.

(89) FIG. 8D illustrates an example of displaying an image using a pulsed light beam and a Lissajous scanning pattern **860**. In one example, Lissajous scanning pattern **860** may be implemented using bi-resonant MEMS beam steering system. As scanning patterns S1-S8, Lissajous scanning pattern **860** may also be independent of the content to be displayed. FIG. 8D also shows a zoom-in view of a subset of equally spaced pixels. A laser pulse may be emitted when the laser beam is steered to a direction corresponding to a pixel **862** that may need to be bright, and the laser beam may be turned off when the laser beam is steered to a direction corresponding to a pixel **862** that may need to be dark. The laser beam dwell time (e.g., pulse duration) at each pixel may be constant, whether the displayed image has a low or high fill factor for bright pixels.

(90) As described above, in “raster” scanning or “Lissajous” scanning, high-speed laser modulation

may be used to generate short laser pulses to generate bright pixels, thereby projecting arbitrary pixel content. A bright pixel may be displayed to the user when the laser emits a short laser pulse while the scanner points to a corresponding direction for the pixel. The brightness of each bright pixel may be limited due to the short dwell time and short laser pulse for each pixel (e.g., less than about 20 ns or less than about 10 ns), whether the displayed image has a low or high fill factor for bright pixels. To achieve a high brightness, the short-pulsed laser beam may need to have high peak and average optical power. However, higher electrical driving power may be needed to achieve higher optical power (or brightness) in these short-pulsed laser sources. Furthermore, in waveguide-based head-mounted display systems, such as head-mounted display system 700, the optical efficiencies from the light source, through the input coupler, waveguide, and output coupler(s), to the eyepiece may be low, such as about 1% or a few percents, due to, for example, low efficiencies of the input and output couplers. Due to the lower optical efficiencies of waveguide-based head-mounted display systems, higher optical output power from the laser sources may be needed to achieve the desired brightness in the images displayed to the eyepiece of the waveguide-based head-mounted display systems. To achieve the higher optical output power, the head-mounted display systems may need to have high electrical driving power. In addition, the optical systems of the light beam scanning-based head-mounted display systems may need to have high damage thresholds in order to manipulate the high-power short-pulse laser beams without being damaged.

(91) In many artificial reality applications, such as many augmented reality displays, the images to be displayed may have low fill factors, where the content may be displayed using only a small portion of the available pixels for each image frame. However, the “raster” or “Lissajous” scanning techniques may still need to scan to all pixel directions, and the light beam dwell time for each pixel may be constant and short, because the scanning path is fixed and is content independent. Therefore, to achieve the desired image brightness, the short-pulsed laser beams may need to have high peak power, and thus the head-mounted display systems may need to have high electrical peak power and high laser damage thresholds. In addition, scanning to corresponding directions for all pixel regardless of the content to be displayed may also consume more electrical power for driving the scanning elements (e.g., MEMS mirrors) of the head-mounted display systems.

(92) According to certain embodiments disclosed herein, vector scanning techniques may be used in some head-mounted display systems (e.g., some augmented reality or mixed reality display systems that include waveguide displays) to steer a light beam according to a scan path that may be content dependent, where the light beam may not need to be steered into all directions for displaying most 2D images. In vector scanning, the content to be displayed may be defined using coordinates and lines, rather than pixels in a pixel grid, where the light beam may function as a paintbrush and may be steered by a beam steering device according to a scan path to paint an image. The beam steering device may be controlled to steer light according to an arbitrary content-dependent scan path, rather than according to a fixed scan path as in “raster” scanning or “Lissajous” scanning. The scan path may be determined based on the content to be displayed. To display different content, different vector scan paths may be used. The color, brightness, pulse duration, and the width of the light beam may be controlled during the scanning process to generate the desired display content.

(93) In some examples, the scan path may be optimized to reduce the overall length of the scan path. The light beam may not need to be turned on and off for each pixel direction, and can remain on when the light beam is steered along a straight or curved line. The light source can be turned off and the steering can be at a faster speed during transitions between content of the image, thereby reducing power consumption and increasing the dwell time for content to be displayed. Therefore, compared with “raster” scanning or “Lissajous” scanning techniques, vector scanning techniques disclosed herein may display an image using a content-dependent scan path that may minimize the scan path length and reduce the power used for the scanning. Because the scan path length may be

reduced and the light beam may not need to be steered into all directions for displaying a 2D image, the effective dwell time of the light beam at each steering direction can be increased to increase the brightness of each image frame. Therefore, to achieve the desired brightness of the displayed image, the optical output power of the light source can be reduced. For example, to display a low-fill image that includes only one line at a frame rate of 60 Hz and an effective resolution of 1000×1000 pixels, the dwell time at each scanning direction can be about 16.67 μs (1/60/1000), rather than about 16.67 ns as in raster scanning or “Lissajous” scanning. As such, with the same optical output power of the light beam, the brightness of the content displayed using vector scanning can be about 1000 times of that of the content displayed using raster scanning or “Lissajous” scanning. Alternatively, compared with raster scanning or “Lissajous” scanning, the same or higher brightness can be achieved by vector scanning using a light beam with a much lower power.

(94) Furthermore, in some examples, the scan path can be modified to include curves or loops (rather than sharp turns) in the scan path. Using curves or loops (rather than sharp turns) may reduce the electrical power used for steering the light beam. For example, in MEMS vector scanner, there may be a relationship between the torque needed to actuate (e.g., rotate) the MEMS mirror and the drive power. The scan path may be optimized to maintain sharp corners in the content while reducing the electrical power used for steering the light beam, for example, by adding a loop at a sharp corner on the scan path and turning off the illumination source during the steering according to the loop. Using curves or loops (rather than sharp turns) in the scan path can reduce the torque for actuating the MEMS mirror, and thus may reduce the electrical power for actuating the MEMS mirror. Using curves or loops at sharp corners may also enable the scanner to scan at a higher speed at the sharp corners.

(95) FIG. 9 illustrates an example of a vector scanning-based head-mounted display system **900** according to certain embodiments. In the illustrated example, head-mounted display system **900** may include a processing unit **910**, an illumination source **920**, a vector scanner **930**, and a waveguide display **940**. Processing unit **910** may include one or more general-purpose or special-purpose processors, and may determine the images to be displayed, determine a scanning pattern for controlling vector scanner **930**, determine the modulation timing for modulating illumination source **920**, and generate control signals for controlling vector scanner **930** and illumination source **920**.

(96) In the example illustrated in FIG. 9, processing unit **910** may determine an image to be displayed for an image frame by executing a functional block **912**. The image to be displayed may be determined based on, for example, an object in the ambient environment of head-mounted display system **900**, to provide additional information regarding the object. In another example, the image to be displayed may include information requested by the user or information received by head-mounted display system **900** from another user device (e.g., a cellphone, a medical device, a computer, etc.). In another example, the image to be displayed may be determined based on the gazing direction of the user or a gesture of the user of head-mounted display system **900**, to display a computer generated graph for immersive augmented reality applications (e.g., a video game).

(97) Based on the content of the image to be displayed, processing unit **910** may implement a scanning algorithm by executing a functional block **914** to determine a vector scanning pattern for displaying the image. For example, functional block **914** may be executed to determine coordinates and scan lines (straight and/or curved) for painting or drawing an image. In one example, the scan lines may be determined by tracing edges of the content. In some examples, high-fill content may be converted to low-fill content, for example, using contours or outlines of the high-fill content. In some examples, a high-fill area of the content may be displayed using regional raster scanning to fill the area, while low-fill areas of the content may be displayed using vector scanning.

(98) Processing unit **910** may also execute a functional block **916** to determine a vector scanning pattern by, for example, determining a sequence for traversing the scan lines and determining

transition lines for connecting unconnected scan lines. The transition lines may be straight and/or curved. The vector scanning pattern may be optimized to reduce the length of the transition lines and reduce the total scan path length. Based on the vector scanning pattern, processing unit **910** may also determine the timing information for modulating illumination source **920** to emit light in appropriate time periods during the scanning process, such as the time for turning on illumination source **920**, the duration of each light pulse, the time for turning off illumination source **920**, the output power of illumination source **920** as a function of time, the scanning speed for each scan line, and the like. For example, the scanning speed for a scan line may be reduced when the content corresponding to the scan line needs to have higher brightness.

(99) In some examples, processing unit **910** may execute a functional block **918** to generate control signals for controlling illumination source **920** and vector scanner **930**, based on the vector scanning pattern and the timing information for modulated illumination source **920** determined using functional block **916**. In some examples, the control signals output by processing unit **910** may be digital signals that may be converted to analog signals for modulating illumination source **920** or driving actuator(s) for rotating one or more mirrors (e.g., MEMs mirrors) of vector scanner **930**. The control signals for modulating illumination source **920** (e.g., source modulation signals) and the control signals for controlling vector scanner **930** (e.g., steering control signals) may be synchronized so that illumination source **920** may emit light when vector scanner **930** directs the light beam to directions corresponding to the scan lines.

(100) Illumination source **920** may be controlled by the source modulation signals to emit light during different time periods, thereby generating light pulses having different durations, intensities, and intervals. The emitted light pulses may be directed by vector scanner **930** to appropriated directions based on the steering control signals for controlling vector scanner **930**. The light pulses having different propagation directions may be coupled into the waveguide (e.g., a substrate) of waveguide display **940**, guided by the waveguide, and coupled out of the waveguide towards the eyebox, as described above with respect to, for example, FIG. 7. Waveguide display **940** may be similar to waveguide display **730**, and may be transparent to visible light so as to allow the user to view the ambient environment by seeing through waveguide display **940**.

(101) FIGS. **10A-10D** illustrate examples of displaying an image using vector scanning according to certain embodiments. FIG. **10A** shows an image **1000** to be displayed by a light beam scanning-based head-mounted display system disclosed herein, such as head-mounted display system **900**. In the illustrated example, image **1000** may include low-fill content, such as text “2+2.” Image **1000** may be drawn using several lines, such as a curved line **1010** and a straight line **1012** for showing a number “2,” a horizontal line **1020** and a vertical line **1022** for showing a symbol “+,” and a curved line **1030** and a straight line **1032** for showing another number “2.” Therefore, image **100** may be displayed by steering a light beam along the six scan lines, where light may only be emitted when the scanner steers a light beam along the six scan lines. In this way, the scanning path length can be reduced and the dwell time of the light beam at each location on the scan lines can be increased. Therefore, a high brightness may be achieved using a light beam that may have a relatively low optical power.

(102) FIG. **10B** illustrates an example of a vector scanning pattern **1002** for displaying image **1000** according to certain embodiments. Vector scanning pattern **1002** may include scan lines **1014**, **1016**, **1024**, **1028**, **1036**, and **1038**. The scan lines may be connected directly or by straight transition lines **1018**, **1026**, and **1034**. Light may be emitted when the scanner steers the light beam to move a light spot along the scan lines, and may be turned off when the scanner scans along the transition lines. In some examples, the speed for scanning along the transition lines may be higher than the speed for scanning along the scan lines, thereby increasing the time that the light may be emitted during a frame period. The sequence of traversing the scan lines and the transition lines in the scanning pattern may be selected to reduce the total length of the transition lines. In one example, after scanning along scan line **1016**, the light beam may be moved along a transition line

1021 to point to the bottom of scan line **1028**, and may be moved long scan line **1028** to the top of scan line **1028**. The light beam may then be moved along a transition line **1025** to point to the left end of scan line **1024**, and may be moved along scan line **1024** to the right end of scan line **1024**. The light beam may then be pointed to the start of scan line **1036**. Such a scanning pattern may reduce the total length of the transition lines and the total scan path length of the scanning pattern. (103) FIG. **10C** illustrates another example of a vector scanning pattern **1004** for displaying image **1000** according to certain embodiments. Vector scanning pattern **1004** may be similar to vector scanning pattern **1002**, but may include curved transition lines **1040**, **1042**, and **1044** between points of the scan lines that are not connected. Vector scanning pattern **1004** may be used for controlling scanners with beam steering mechanisms that may not make sharp turns at high speeds due to certain inertia and/or mechanical limitations or may consume more power for making sharp turns. For example, for some scanners, making sharp turns or changes of direction may need to be performed at a lower speed, which may reduce the overall scanning speed. For some scanners, such as some MEMS vector scanners, a higher torque may be needed in order to make sharp turns or changes of direction, and applying a higher torque may use more electrical power. Using the curved transition lines as shown in FIG. **10C**, the scanning speed for the transition lines can be higher, and the electrical power used to steer the beam along the transition lines can be lower.

(104) FIG. **10D** illustrates yet another example of a vector scanning pattern **1006** for displaying image **1000** according to certain embodiments. Vector scanning pattern **1006** may be similar to vector scanning pattern **1004**, but may include additional loops **1050** and **1052** between scan lines that may be connected but may have a small angle (less than 90°) between them. For example, scan lines **1014** and **1016** may be connected, but transition from scan line **1014** to scan line **1016** may include making a sharp turn, which may need to be performed at a lower speed and/or may need a higher torque and a higher electrical power. Including a loop **1050** between scan lines **1014** and **1016** may enable the scanning at a relative high speed with smaller torque (and lower power) for rotating the scanner.

(105) Using the vector scanning patterns and vector scanners disclosed herein may increase the brightness of the displayed image, without using light beams with higher optical power. In one example, to project a full-fill white image with one million pixels in a half-angle circular field of view about 10° using an example of a 2D raster scanning display and a waveguide display with an overall optical efficiency about 1% and an eyebox size of 10 mm×10 mm at 100 nits, the input power to the waveguide may need to be approximately $100 \text{ nits} \times (0.010 \text{ m} \times 0.010 \text{ m}) \times 4\pi \times \sin^2(10^\circ) / 1\% = 0.38 \text{ lumens}$. In contrast, to display a low-fill image including about 1000 effective pixels using the same waveguide display and a vector scanner with an 80% drawing duty cycle (e.g., due to gaps between line content), a brightness of approximately $100 \text{ nits} \times 1,000,000 \text{ pixels} / 1000 \text{ pixels} \times 80\% = 80,000 \text{ nits}$ may be achieved with the same equivalent input optical of about 0.38 lumens. To achieve the same 100 nits brightness for the content including about 1000 effective pixels using the vector scanner, the input power to the waveguide may only need to be approximately $0.38 \text{ lumens} \times 1000 \text{ pixels} / 1,000,000 \text{ pixels} / 80\% = 474 \text{ microlumens}$.

(106) In some vector scanning-based head-mounted display systems, the light beam can be controlled to have different beam widths during the scanning process, such that the lines of the image generated in a single scan path may have different thicknesses. The brightness of the portion of content generated using light beams with the larger beam widths can be maintained at a high level using the same output power from the light source, for example, by reducing the scanning speed for generating the portion of content.

(107) FIG. **11A** illustrates an example of a vector scanning-based head-mounted display system **1100** with a variable beam width according to certain embodiments. Head-mounted display system **1100** may be similar to head-mounted display system **900**, and may include additional dynamic blur optics **1110** that may be controlled to dynamically change the beam width of the light beam

emitted by illumination source **920**. For example, when the content to be displayed may be represented by lines of different thickness, processing unit **910** may determine the thicknesses of the scan lines and thus the blur needed to generate the scan lines of different thicknesses, and may also generate blurring control signals to control dynamic blur optics **1110**. Dynamic blur optics may control the beam width of the light beam to be steered by vector scanner **930**. Dynamic blur optics may include, for example, a dynamic lens such as a liquid lens (e.g., a liquid crystal lens), a spatial light modulator (SLM) with a variable computer generated hologram (CGH) pattern, a dynamic zooming lens, or some other optical element capable of controlling the beam width of the transmitted light beam. In some examples, dynamic blur optics **1110** may be part of illumination source **920**. In some examples, dynamic blur optics **1110** may be positioned between vector scanner **930** and waveguide display **940**. A larger beam width may allow more effective pixels to be covered in a single scan line, thereby reducing the number of scan lines for displaying some high-fill regions of the content.

(108) FIG. **11B** illustrates an example of a scanning pattern **1102** for displaying an image using vector scanning-based head-mounted display system **1100** according to certain embodiments. In the example illustrated in FIG. **11B**, the image to be displayed may be similar to image **1000**, but the symbol “+” may have a higher line width than the numbers (e.g., “2”). In another example, letters in the content of the image may include letters of different font sizes and/or styles (e.g., regular, bold, or italic fonts). If a light beam having a same beam width is used, the symbol “+” or the larger or bold fonts may need to be displayed using multiple scan lines. When dynamic blur optics such as dynamic blur optics **1110** are used in the head-mounted display system, the content with different line widths may be displayed by dynamically controlling the beam width of the light beam using the dynamic blur optics.

(109) In the example shown in FIG. **11B**, scanning pattern **1102** may include thinner scan lines **1120**, **1124**, **1136**, and **1140**, and thicker scan lines **1128** and **1132**. The scan lines may be connected by transition lines **1122**, **1126**, **1130**, **1134**, and **1138**, as described above with respect to FIGS. **10B-10D**. When the light beam is steered along thinner scan lines **1120**, **1124**, **1136**, and **1140**, the dynamic blur optics may be controlled such that the light beam output by the dynamic blur optics may have a smaller beam width. When the light beam is steered along thicker scan lines **1128** and **1132**, the dynamic blur optics may be controlled such that the light beam output by the dynamic blur optics may have a larger beam width. In this way, the symbol “+” may be displayed by steering the light beam to traverse scan lines **1128** and **1132** once.

(110) FIGS. **12A-12C** illustrate an example of a method of displaying an image by vector scanning according to certain embodiments. In the illustrated example, a high-fill image **1200** may be converted into to a low-fill image **1204** that may be represented by a plurality of lines, where the low-fill image **1204** may be displayed using the vector scanning techniques and systems disclosed herein.

(111) In the example illustrated in FIGS. **12A-12C**, high-fill image **1200** shown in FIG. **12A** may first be converted into a low-fill image **1202** that can be represented by a plurality of lines as shown in FIG. **12B**. For example, techniques such as edge detection techniques may be used to detect edges of objects in image **1200**. The detected edges may be represented by the plurality of lines. In some examples, one or more iterations of the image conversion may be performed to further reduce the number of lines used to represent features in image **1200**, while maintaining the lines for more relevant features, as shown in FIG. **12C**. In this way, main features of a high-fill image such as image **1200** may be displayed by vector scanning using a small number of scan lines, where the main features may be displayed with high brightness without increasing the power consumption of the head-mounted display system. In some examples, high-fill regions of the image may be displayed using a raster scanning pattern, while other regions of the image may be displayed by a vector scanning pattern. Thus, the image may be displayed using a hybrid scanning pattern.

(112) In some waveguide display systems, the brightness of the output image may vary across the

field of view, the wavelength range, and/or eye positions. The brightness non-uniformity may be caused by the waveguide, the input/output couplers, or components between the light source and the waveguide. Uniformity correction may need to be performed to pre-compensate the light, thereby minimizing perceived non-uniformity by the eye. Uniformity correction may be achieved, for example, by changing the amount of light projected per pixel, such as by driving a micro-LED pixel with signals of different electrical power or varying the amplitude of a pulsed laser in a raster scanning display system. In some examples, the vector scanning-based head-mounted display systems disclosed herein can additionally or alternatively perform brightness uniformity correction by changing the linear and/or angular velocity of the vector scanner. As such, uniformity correction may be achieved in the vector scanning-based head-mounted display systems disclosed herein by controlling the light source modulation, the speed of the vector scanner, or a combination thereof.

(113) FIG. 13 illustrates an example of a waveguide display 1300 according to certain embodiments. Waveguide display 1300 may be similar to waveguide display 730 and waveguide display 940, and may include a waveguide 1310 (e.g., a substrate), one or more input couplers 1320, and one or more output couplers 1322 as described above. As shown in FIG. 13, the brightness of the displayed image at different regions of the eyebox may not be the same. For example, a pupil replicated by waveguide display 1300 at a position A of the eyebox may have a higher brightness than a pupil replicated by waveguide display 1300 at a position B of the eyebox. Therefore, the image perceived by a user's eye at position A and the image perceived by the user's eye at position B may have different brightnesses. The brightness of the image can also be nonuniform for different FOVs and/or different colors. The brightnesses nonuniformity may be caused by, for example, the nonuniform coupling efficiency of input couplers 1320 for different colors and/or different FOVs, the nonuniformity of the coupling efficiency of output couplers 1322 for different colors and/or different FOVs, the nonuniformity of the light intensity of the light beam guided by waveguide 1310 at different regions of output couplers 1322 (e.g., as portions of the guided light being coupled out of the waveguide by output couplers 1322 at different regions of waveguide 1310), the light engine (e.g., including the light source, the modulator, the scanner, the dynamic blur optics, etc.), and the like. It is generally desirable to minimize the brightness nonuniformity perceived by the user's eye.

(114) According to certain embodiments disclosed herein, the brightness of the generated image may be pre-compensated during the scanning process, for example, by controlling the output power of the light source (e.g., by controlling the modulation of the light source), and/or the scanning speed (e.g., linear or angular velocity) and thus the dwell time at each scanning direction, thereby improving the brightness uniformity of the image output from a waveguide display to the eyebox. In some examples, the brightness uniformity correction can be made across the eye box (independent of the eye pupil position), such that the user's eye may perceive the same brightness regardless of the location of user's eye. In some examples, the brightness uniformity correction can be made based on the location of the user's eye pupil (and the pupil location-dependent field of view), and pre-determined nonuniformity information of the display system for different eyebox locations, FOVs, and/or colors, where the location of the user's eye pupil may be detected by an eye-tracking system (e.g., including an infrared light emitter and a camera) described with respect to, for example, FIG. 1.

(115) FIG. 14A illustrates an example of a vector scanning-based head-mounted display system 1400 capable of non-uniformity correction according to certain embodiments. Head-mounted display system 1400 may be similar to head-mounted display system 900 or 1100. In head-mounted display system 1400, waveguide display nonuniformity data 1410 may be pre-determined and saved to a memory device. Processing unit 910, when executing functional block 916, may read waveguide display nonuniformity data 1410 from the memory device, and optionally obtaining user pupil location information from an optional eye-tracking subsystem 1420, to determine the scanning pattern, and the corresponding scanning speed (e.g., linear and/or angular velocity) and

light source modulation during the scan path. For example, the scanning speed and/or the modulation of the light source (including the output power and/or the timing such as pulse width/duration of the light source) may be varied to pre-compensate the brightness nonuniformity that may otherwise be caused by illumination source **920**, vector scanner **930**, and/or waveguide display **940**.

(116) FIG. **14B** illustrates an example of non-uniformity correction using vector scanning-based head-mounted display system **1400** of FIG. **14A** according to certain embodiments. In the example shown in FIG. **14B**, the output power of the illumination source may be constant, but the scanning speed of the vector scanner may be varied along a scan path for display a line, to pre-compensate possible nonuniformity of the waveguide display. For example, if the waveguided display may otherwise cause a lower brightness at the middle of the line, the vector scanner may be controlled to scan at a lower speed at the middle of the line and scan at higher speeds at the two ends of the line, such that the image before entering the waveguide display may have a higher brightness at the middle of the line and lower brightnesses at the two ends of the line, and thus the image output by the waveguide display may have a uniform brightness across the line.

(117) FIG. **15** includes a flowchart **1500** illustrating an example of a method of displaying an image using a vector scanning-based head-mounted display system according to certain embodiments. Operations in flowchart **1500** may be performed using, for example, one or more processors or computing systems (e.g., near-eye display **120** or an electronic system **1600** described below). Although flowchart **1500** may describe the operations as a sequential order, some of the operations may be performed in parallel or concurrently. In addition, the order of the operations may be rearranged. The process may have additional operations not included in the flowchart. Some operations may be optional or may be omitted in some implementations. Some operations may be performed more than one time. Furthermore, embodiments of the methods may be implemented using hardware, software, firmware, middleware, microcode, hardware description languages, or any combination thereof. One of ordinary skill in the art would recognize many variations, modifications, and alternatives.

(118) Operations at block **1510** of flowchart **1500** may include determining, based on content of an image to be displayed by a head-mounted display system, a vector scanning pattern for displaying the image using a scanner. The scanner may include, for example, a MEMS scanner including a micro-mirror and an actuator for rotating the micro-mirror. The vector scanning pattern may include a plurality of scan lines and one or more transition lines connecting the plurality of scan lines. At least one transition line of the one or more transition lines may be a curved line or include a curved section. In one example, the one or more transition lines may include a transition line that may form a loop to connect two scan lines of the plurality of scan lines. As described above, by using the curved transition lines or loops (rather than sharp turns) for connecting the scan lines, the torque and electrical power for rotating the micro-mirror may be reduced, and the scanning speed (e.g., angular velocity) may not need to be reduced to transition from one scan line to another scan line.

(119) In some examples, the content of the image may be determined by one or more processors based on, for example, an object in the ambient environment or an eye gazing direction determined by an eye-tracking system. In some examples, the vector scanning pattern may be determined by detecting edges of features in the image, and determining the vector scanning pattern based on the detected edges of the features in the image. In some examples, the vector scanning pattern may be selected to minimize a total length of the scan path of the vector scanning pattern. In some examples, the vector scanning pattern may include a region that includes a regional raster scanning pattern.

(120) Operations at block **1520** may include generating control signals for controlling a scanner to steer, according to the vector scanning pattern, a light beam emitted by a light source towards a waveguide display of the head-mounted display system. The light source may include a laser or

another light emitter. The scanner may include, for example, a micro-mirror and an actuator for rotating the micro-mirror, such as a MEMS scanner. The waveguide display may include a substrate transparent to visible light, an input coupler configured to couple the light beam into the substrate such that the light beam is guided by the substrate (e.g., through total internal reflection), and an output coupler configured to couple the light beam out of the substrate at one or more locations of the substrate. Controlling the scanner to steer the light beam according to the vector scanning pattern may include controlling the scanner to steer the light beam at a first velocity when the scanner operates according to a scan line of the plurality of scan lines, and steer at a second velocity that is higher than the first velocity when the scanner operates according to a transition line of the one or more transition lines. In some examples, controlling the scanner to steer the light beam according to the vector scanning pattern may include controlling the scanner to steer the light beam at a variable velocity based on brightness information of the image when the scanner operates according to a scan line of the plurality of scan lines. In some examples, controlling the scanner to steer the light beam according to the vector scanning pattern may include generating, based on the brightness nonuniformity information of the waveguide display, control signals for varying a scanning velocity of the scanner, thereby pre-compensating the brightness nonuniformity of the waveguide display.

(121) Operations at block **1530** may include generating modulation signals for modulating the light source such that the light beam is turned on when the scanner operates according to the plurality of scan lines and is turned off when the scanner operates according to the one or more transition lines. In some examples, the light source may include a laser that can be modulated or may include an optical modulator configured to modulate the light beam output by the laser. In some examples, controlling the scanner to steer the light beam according to the vector scanning pattern may include generating, based on the brightness nonuniformity information of the waveguide display, modulation signals for modulating an output power of the light source, thereby pre-compensating the brightness nonuniformity of the waveguide display.

(122) Optional operations at block **1540** may include generating, based on the content of the image, control signals for controlling dynamic blur optics to dynamically change a beam width of the light beam, such that the image may include lines of different thicknesses. In this way, the scan path length can be further reduced. When the beam width of the light beam is changed, the scanning speed may be changed accordingly to achieve the desired brightness of the lines of different thicknesses. The dynamic blur optics may include, for example, a dynamic lens such as a liquid lens, a spatial light modulator (SLM) with a variable computer generated hologram (CGH) pattern, a dynamic zooming lens, or some other optical element capable of controlling the beam width of the transmitted light beam.

(123) Embodiments disclosed herein may be used to implement components of an artificial reality system or may be implemented in conjunction with an artificial reality system. Artificial reality is a form of reality that has been adjusted in some manner before presentation to a user, which may include, for example, a virtual reality, an augmented reality, a mixed reality, a hybrid reality, or some combination and/or derivatives thereof. Artificial reality content may include completely generated content or generated content combined with captured (e.g., real-world) content. The artificial reality content may include video, audio, haptic feedback, or some combination thereof, and any of which may be presented in a single channel or in multiple channels (such as stereo video that produces a three-dimensional effect to the viewer). Additionally, in some embodiments, artificial reality may also be associated with applications, products, accessories, services, or some combination thereof, that are used to, for example, create content in an artificial reality and/or are otherwise used in (e.g., perform activities in) an artificial reality. The artificial reality system that provides the artificial reality content may be implemented on various platforms, including an HMD connected to a host computer system, a standalone HMD, a mobile device or computing system, or any other hardware platform capable of providing artificial reality content to one or more viewers.

(124) FIG. 16 is a simplified block diagram of an example electronic system **1600** of an example near-eye display (e.g., HMD device) for implementing some of the examples disclosed herein. Electronic system **1600** may be used as the electronic system of an HMD device or other near-eye displays described above. In this example, electronic system **1600** may include one or more processor(s) **1610** and memory **1620**. Processor(s) **1610** may be configured to execute instructions for performing operations at a number of components, and can be, for example, a general-purpose processor or microprocessor suitable for implementation within a portable electronic device. Processor(s) **1610** may be communicatively coupled with a plurality of components within electronic system **1600**. To realize this communicative coupling, processor(s) **1610** may communicate with the other illustrated components across a bus **1640**. Bus **1640** may be any subsystem adapted to transfer data within electronic system **1600**. Bus **1640** may include a plurality of computer buses and additional circuitry to transfer data.

(125) Memory **1620** may be coupled to processor(s) **1610**. In some embodiments, memory **1620** may offer both short-term and long-term storage and may be divided into several units. Memory **1620** may be volatile, such as static random access memory (SRAM) and/or dynamic random access memory (DRAM) and/or non-volatile, such as read-only memory (ROM), flash memory, and the like. Furthermore, memory **1620** may include removable storage devices, such as secure digital (SD) cards. Memory **1620** may provide storage of computer-readable instructions, data structures, programs, and other data for electronic system **1600**.

(126) In some embodiments, memory **1620** may store a plurality of applications **1622** through **1624**, which may include any number of applications. Examples of applications may include gaming applications, conferencing applications, video playback applications, or other suitable applications. The applications may include a depth sensing function or eye tracking function. Applications **1622-1624** may include particular instructions to be executed by processor(s) **1610**. In some embodiments, certain applications or parts of applications **1622-1624** may be executable by other hardware **1680**. In certain embodiments, memory **1620** may additionally include secure memory, which may include additional security controls to prevent copying or other unauthorized access to secure information.

(127) In some embodiments, memory **1620** may include an operating system **1625** loaded therein. Operating system **1625** may be operable to initiate the execution of the instructions provided by applications **1622-1624** and/or manage other hardware **1680** as well as interfaces with a wireless communication subsystem **1630** which may include one or more wireless transceivers. Operating system **1625** may be adapted to perform other operations across the components of electronic system **1600** including threading, resource management, data storage control and other similar functionality.

(128) Wireless communication subsystem **1630** may include, for example, an infrared communication device, a wireless communication device and/or chipset (such as a Bluetooth® device, an IEEE 802.11 device, a Wi-Fi device, a WiMax device, cellular communication facilities, etc.), and/or similar communication interfaces. Electronic system **1600** may include one or more antennas **1634** for wireless communication as part of wireless communication subsystem **1630** or as a separate component coupled to any portion of the system. Depending on desired functionality, wireless communication subsystem **1630** may include separate transceivers to communicate with base transceiver stations and other wireless devices and access points, which may include communicating with different data networks and/or network types, such as wireless wide-area networks (WWANs), wireless local area networks (WLANs), or wireless personal area networks (WPANs). A WWAN may be, for example, a WiMax (IEEE 802.16) network. A WLAN may be, for example, an IEEE 802.11x network. A WPAN may be, for example, a Bluetooth network, an IEEE 802.15x, or some other types of network. The techniques described herein may also be used for any combination of WWAN, WLAN, and/or WPAN. Wireless communications subsystem **1630** may permit data to be exchanged with a network, other computer systems, and/or any other devices

described herein. Wireless communication subsystem **1630** may include a means for transmitting or receiving data, such as identifiers of HMD devices, position data, a geographic map, a heat map, photos, or videos, using antenna(s) **1634** and wireless link(s) **1632**.

(129) Embodiments of electronic system **1600** may also include one or more sensors **1690**.

Sensor(s) **1690** may include, for example, an image sensor, an accelerometer, a pressure sensor, a temperature sensor, a proximity sensor, a magnetometer, a gyroscope, an inertial sensor (e.g., a that combines an accelerometer and a gyroscope), an ambient light sensor, or any other similar operable to provide sensory output and/or receive sensory input, such as a depth sensor or a position sensor.

(130) Electronic system **1600** may include a display **1660**. Display **1660** may be a near-eye display, and may graphically present information, such as images, videos, and various instructions, from electronic system **1600** to a user. Such information may be derived from one or more applications **1622-1624**, virtual reality engine **1626**, one or more other hardware **1680**, a combination thereof, or any other suitable means for resolving graphical content for the user (e.g., by operating system **1625**). Display **1660** may use LCD technology, LED technology (including, for example, OLED, ILED, μ -LED, AMOLED, TOLED, etc.), light emitting polymer display (LPD) technology, or some other display technology.

(131) Electronic system **1600** may include a user input/output **1670**. User input/output **1670** may allow a user to send action requests to electronic system **1600**. An action request may be a request to perform a particular action. For example, an action request may be to start or end an application or to perform a particular action within the application. User input/output **1670** may include one or more input devices. Example input devices may include a touchscreen, a touch pad, microphone(s), button(s), dial(s), switch(es), a keyboard, a mouse, a game controller, or any other suitable device for receiving action requests and communicating the received action requests to electronic system **1600**. In some embodiments, user input/output **1670** may provide haptic feedback to the user in accordance with instructions received from electronic system **1600**. For example, the haptic feedback may be provided when an action request is received or has been performed.

(132) Electronic system **1600** may include a camera **1650** that may be used to take photos or videos of a user, for example, for tracking the user's eye position. Camera **1650** may also be used to take photos or videos of the environment, for example, for VR, AR, or MR applications. Camera **1650** may include, for example, a complementary metal-oxide-semiconductor (CMOS) image sensor with a few millions or tens of millions of pixels. In some implementations, camera **1650** may include two or more cameras that may be used to capture 3-D images.

(133) In some embodiments, electronic system **1600** may include a plurality of other hardware **1680**. Each of other hardware **1680** may be a physical within electronic system **1600**. While each of other hardware **1680** may be permanently configured as a structure, some of other hardware **1680** may be temporarily configured to perform specific functions or temporarily activated. Examples of other hardware **1680** may include, for example, an audio output and/or input (e.g., a microphone or speaker), a near field communication (NFC), a rechargeable battery, a battery management system, a wired/wireless battery charging system, etc. In some embodiments, one or more functions of other hardware **1680** may be implemented in software.

(134) In some embodiments, memory **1620** of electronic system **1600** may also store a virtual reality engine **1626**. Virtual reality engine **1626** may execute applications within electronic system **1600** and receive position information, acceleration information, velocity information, predicted future positions, or any combination thereof of the HMD device from the various sensors. In some embodiments, the information received by virtual reality engine **1626** may be used for producing a signal (e.g., display instructions) to display **1660**. For example, if the received information indicates that the user has looked to the left, virtual reality engine **1626** may generate content for the HMD device that mirrors the user's movement in a virtual environment. Additionally, virtual reality engine **1626** may perform an action within an application in response to an action request received from user input/output **1670** and provide feedback to the user. The provided feedback may

be visual, audible, or haptic feedback. In some implementations, processor(s) **1610** may include one or more GPUs that may execute virtual reality engine **1626**.

(135) The methods, systems, and devices discussed above are examples. Various embodiments may omit, substitute, or add various procedures or components as appropriate. For instance, in alternative configurations, the methods described may be performed in an order different from that described, and/or various stages may be added, omitted, and/or combined. Also, features described with respect to certain embodiments may be combined in various other embodiments. Different aspects and elements of the embodiments may be combined in a similar manner. Also, technology evolves and, thus, many of the elements are examples that do not limit the scope of the disclosure to those specific examples.

(136) Specific details are given in the description to provide a thorough understanding of the embodiments. However, embodiments may be practiced without these specific details. For example, well-known circuits, processes, systems, structures, and techniques have been shown without unnecessary detail in order to avoid obscuring the embodiments. This description provides example embodiments only, and is not intended to limit the scope, applicability, or configuration of the invention. Rather, the preceding description of the embodiments will provide those skilled in the art with an enabling description for implementing various embodiments. Various changes may be made in the function and arrangement of elements without departing from the spirit and scope of the present disclosure.

(137) Also, some embodiments were described as processes depicted as flow diagrams or block diagrams. Although each may describe the operations as a sequential process, many of the operations may be performed in parallel or concurrently. In addition, the order of the operations may be rearranged. A process may have additional steps not included in the figure. Furthermore, embodiments of the methods may be implemented by hardware, software, firmware, middleware, microcode, hardware description languages, or any combination thereof.

(138) Terms, “and” and “or” as used herein, may include a variety of meanings that are also expected to depend at least in part upon the context in which such terms are used. Typically, “or” if used to associate a list, such as A, B, or C, is intended to mean A, B, and C, here used in the inclusive sense, as well as A, B, or C, here used in the exclusive sense. In addition, the term “one or more” as used herein may be used to describe any feature, structure, or characteristic in the singular or may be used to describe some combination of features, structures, or characteristics. However, it should be noted that this is merely an illustrative example and claimed subject matter is not limited to this example. Furthermore, the term “at least one of” if used to associate a list, such as A, B, or C, can be interpreted to mean A, B, C, or any combination of A, B, and/or C, such as AB, AC, BC, AA, ABC, AAB, AABBBCCC, etc.

(139) The specification and drawings are, accordingly, to be regarded in an illustrative rather than a restrictive sense. It will, however, be evident that additions, subtractions, deletions, and other modifications and changes may be made thereunto without departing from the broader spirit and scope as set forth in the claims. Thus, although specific embodiments have been described, these are not intended to be limiting. Various modifications and equivalents are within the scope of the following claims.

Claims

1. A head-mounted display system comprising: a light source configured to emit a light beam; a scanner configured to steer the light beam; a waveguide display configured to receive the light beam from the scanner and direct the light beam towards an eyebox of the head-mounted display system; one or more processors electrically coupled to the light source and the scanner; and one or more memory devices storing instructions that, when executed by the one or more processors, cause the one or more processors to perform operations including: determining, based on content of

an image, a vector scanning pattern for displaying the image, wherein the vector scanning pattern includes a plurality of scan lines and one or more transition lines connecting the plurality of scan lines, and wherein at least one transition line of the one or more transition lines includes a curved section; generating control signals for controlling the scanner to steer the light beam according to the vector scanning pattern; and generating modulation signals for modulating the light source such that the light beam is turned on when the scanner operates according to the plurality of scan lines and is turned off when the scanner operates according to the one or more transition lines.

2. The head-mounted display system of claim 1, wherein the one or more transition lines include a transition line that forms a loop to connect two scan lines of the plurality of scan lines.

3. The head-mounted display system of claim 1, wherein controlling the scanner to steer the light beam according to the vector scanning pattern comprises controlling the scanner to: steer the light beam at a first velocity when the scanner operates according to a scan line of the plurality of scan lines; and steer at a second velocity that is higher than the first velocity when the scanner operates according to a transition line of the one or more transition lines.

4. The head-mounted display system of claim 1, wherein controlling the scanner to steer the light beam according to the vector scanning pattern comprises controlling the scanner to steer the light beam at a variable velocity based on brightness information of the image when the scanner operates according to a scan line of the plurality of scan lines.

5. The head-mounted display system of claim 1, further comprising dynamic blur optics configured to dynamically change a beam width of the light beam emitted by the light source.

6. The head-mounted display system of claim 5, wherein the instructions further cause the one or more processors to control the dynamic blur optics based on the content of the image to dynamically change the beam width of the light beam emitted by the light source.

7. The head-mounted display system of claim 1, wherein: the one or more memory devices further store brightness nonuniformity information of the waveguide display; and the instructions further cause the one or more processors to compensate brightness nonuniformity of the waveguide display by performing at least one of: generating, based on the brightness nonuniformity information of the waveguide display, control signals for varying a scanning velocity of the scanner; or generating, based on the brightness nonuniformity information of the waveguide display, modulation signals for modulating an output power of the light source.

8. The head-mounted display system of claim 1, wherein the scanner includes a micro-mirror and an actuator for rotating the micro-mirror.

9. The head-mounted display system of claim 1, wherein the light source includes at least one of a laser that can be modulated or an optical modulator configured to modulate the light beam.

10. The head-mounted display system of claim 1, wherein the waveguide display includes: a substrate transparent to visible light; an input coupler configured to couple the light beam into the substrate such that the light beam is guided by the substrate; and an output coupler configured to couple the light beam out of the substrate at one or more locations of the substrate.

11. The head-mounted display system of claim 1, further comprising an eye-tracking system.

12. The head-mounted display system of claim 11, wherein the instructions further cause the one or more processors to determine the content of the image based on outputs of the eye-tracking system.

13. A processor-implemented method comprising: determining, based on content of an image to be displayed by a head-mounted display system, a vector scanning pattern for displaying the image, wherein the vector scanning pattern includes a plurality of scan lines and one or more transition lines connecting the plurality of scan lines, and wherein at least one transition line of the one or more transition lines includes a curved section; generating control signals for controlling a scanner to steer, according to the vector scanning pattern, a light beam emitted by a light source towards a waveguide display of the head-mounted display system; and generating modulation signals for modulating the light source such that the light beam is turned on when the scanner operates according to the plurality of scan lines and is turned off when the scanner operates according to the

one or more transition lines.

14. The processor-implemented method of claim 13, wherein the one or more transition lines include a transition line that forms a loop to connect two scan lines of the plurality of scan lines.

15. The processor-implemented method of claim 13, wherein controlling the scanner to steer the light beam according to the vector scanning pattern comprises controlling the scanner to: steer the light beam at a first velocity when the scanner operates according to a scan line of the plurality of scan lines; and steer at a second velocity that is higher than the first velocity when the scanner operates according to a transition line of the one or more transition lines.

16. The processor-implemented method of claim 13, wherein controlling the scanner to steer the light beam according to the vector scanning pattern comprises controlling the scanner to steer the light beam at a variable velocity based on brightness information of the image when the scanner operates according to a scan line of the plurality of scan lines.

17. The processor-implemented method of claim 13, further comprising generating, based on the content of the image, control signals for controlling dynamic blur optics to dynamically change a beam width of the light beam.

18. The processor-implemented method of claim 13, further comprising: obtaining brightness nonuniformity information of the waveguide display; compensating brightness nonuniformity of the waveguide display by performing at least one of: generating, based on the brightness nonuniformity information of the waveguide display, control signals for varying a scanning velocity of the scanner; or generating, based on the brightness nonuniformity information of the waveguide display, modulation signals for modulating an output power of the light source.

19. The processor-implemented method of claim 13, wherein the vector scanning pattern includes a region that includes a raster scanning pattern.

20. The processor-implemented method of claim 13, wherein determining, based on the content of the image to be displayed by the head-mounted display system, the vector scanning pattern for displaying the image comprises: detecting edges of features in the image; and determining the vector scanning pattern based on the detected edges of the features in the image.
